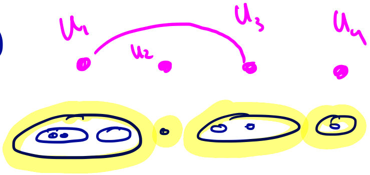
The setting - preliminaries — 1/2

$G = (V, E)$, wt vector w, Assume $\exists$ a perfect matching.

Dual: max: $\sum y_U$

st: $\sum_{U \text{ s.t. } e \in \delta(U)} y_U \leq w_e$, $\forall$ edge e (*)

$y_U \geq 0$, $\forall U$, $|U| \geq 3$ odd



y be a dual feasible solution, with $\Omega = \mathrm{supp}(y)$ will be laminar.

$E_y = \{ e : w_e - \sum_{U, e \in U} y_U = 0 \}$. $G_y = (V, E_y)$

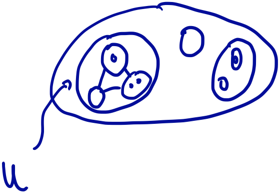
Ω^{max} = inclusionwise maximal elts of Ω - note Ω^{max} is a partition of V

$G' = G_y / \Omega_{max}$, G' has edge (U_i, U_j), $U_i, U_j \in \Omega^{max}$ iff G_y has such an edge.

The setting - preliminaries — 2/2

At each iteration, the alg. maintains and updates

- Ω ; Ω^{max}, Ω laminar.
- y, E_y, G_y, G'
- Hamiltonian cycle $\underline{C_U}$ on the subgraph $\underline{G_y[U] / \text{maximal subsets of } U}$.



- Alg. also maintains a matching on G'

The Algorithm — 1/7

- Initialize with : $\Omega = \Omega^{max} = \{\{v\} : v \in V\}$

  $y_v \equiv 0, \quad M = \phi.$

  ↳ dual feasible ↳ primal infeasible

- Alg does – at high level –
  - Search for a P.M. in G'. → If found, must be opt.

    Build alt trees. If we find augmenting path augment

    if we find blossom, we contract.

    If no P.M., no aug. path, no blossom to ~~shan~~ collapse

    ⇒ update y to improve (increase) dual.

The Algorithm — 2/7

Let X be unmatched vertices of G'

i.e. vertices missed by current M.

<u>Case 1</u>: G' has a M-alternating X-X

walk (of +'ve length) That is an

augmenting path - P -

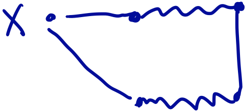
update $M \longleftarrow M \triangle EP$

and iterate.

The Algorithm — 3/7

<u>Case 2:</u> G' has an M-alt $X-X$ walk, that is not an augmenting path.

← case 1.



← <u>Case 2</u>. The walk must contain an M-flower $\Rightarrow$ contains an M-blossom.

Let C denote the blossom — i.e., the odd length cycle

$$u = \bigcup_{v \in C} v.$$

<u>Update</u>

$\Omega \leftarrow \Omega \cup \{u\}$ // shrinking, clearly maintains laminarity.

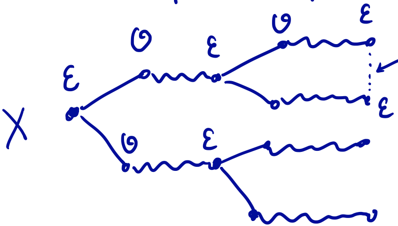
$y_u \leftarrow 0$

$M \leftarrow M \setminus E_C$

$C_u \leftarrow C$

<u>Iterate</u>.

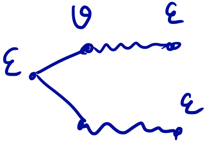
<u>Case 3</u>: G' has no M-alt. $X - X$ walk of positive length.



<u>no</u> $\mathcal{E}$-$\mathcal{E}$ edge o/w we are in <u>case 2</u>.

$\mathcal{E}$ = vertices at even dist. from X

$\mathcal{O}$ = odd



Let $d \in \mathbb{R}^{E'}$, $d_u = \begin{cases} +1 & \text{if } u \in \mathcal{E} \\ -1 & \text{if } u \in \mathcal{O} \end{cases}$ ←

Dual var update: $y \longleftarrow y + \theta d$

for θ as large as possible, maintaining dual feasibility.

The Algorithm — 5/7

M not perfect $\iff$ X not empty $\iff$ $|\underline{\mathcal{E}}| > |\underline{\mathcal{O}}|$

$\Rightarrow$ dual opt. value: $\sum y_u$ increases for $\theta > 0$.

By assumption, $\exists$ perf matching $G = (V, E)$ $\Rightarrow$ primal is feasible hence θ is finite as dual is bdd.

When does θ stop? How big can we take θ? $\leftarrow$ decided by requirement of feasibility.

(A) Some y_u becomes zero, $|u| \geq 3$ for $u \in \mathcal{O}$ (note that y_v can be $-$'ve).

Then: $\Omega \longleftarrow \Omega \setminus \{u\}$ deshrinking.

$M \longleftarrow M \cup \{$perf. matching of $C_u - v$, where v covered by M.

(B) Before some y_u becomes zero, some constraint

$\rightarrow$ $\sum_{\substack{u \\ e \in \delta(u)}} y_u \leq w_e$ becomes tight for some $e \in E \setminus E_y$

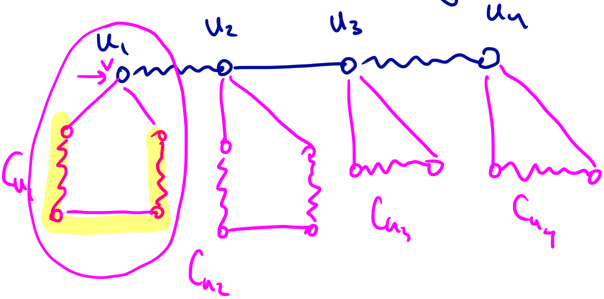
Note: If $e \in E_y$ then constraint is not changed.

|| RP has new edges to work with.

Iterate.

The Algorithm — 6/7

Picture of deshrinking:



Also a picture of how we obtain

a PM in G from our PM in G'

The Algorithm — 7/7

Summary:

- Try to augment.

  If ever perfect ⇒ done.

  Along the way we contract blossoms adding sets to Ω & vars to dual.

- When nothing more to contract, we de-shrink or add new edges